ISSSP 2026 — TRACK 3: LUNAR SURFACE ROBOTICS · FINALS SCORING CRITERION 1

# Community Building & Publicity Campaign Guide

Comprehensive Strategy, Outreach Blueprint, Communications Specialist Playbook, Ready-to-Publish Copy & Team Collaboration Charter (Deadline: 31st Oct — Max Score: 10/10)

| | |
|---|---|
| **Project:** L-MINER Mk-II Autonomous Payload | **Institution:** IST Islamabad |
| **Scoring Item:** Item 1: Community Building (10 Pts) | **Team:** Haider, Wasay, Saad, Hajra |

## 1. Strategic Alignment with Table 6 Finals Scoring Rules

Table 6 allocates 10 points out of 100 to Item 1: Community Building (until 31st October). The scoring criteria are judged by the Organizing Committee and appointed evaluators on two specific dimensions:

- **Publicize Contest Concept (5 Points):** Actively promoting the contest vision, themes, and societal impact across university communities, social channels, and youth STEM forums. Demonstrates public engagement, reach, and advocacy for lunar science.
- **Team Cooperation & Communication (5 Points):** Demonstrating cohesive interdisciplinary teamwork, transparent task allocation, systematic meeting logs, and structured communications leadership led by a designated Communications Specialist.

**Judge Evaluation Mandate:** To secure the maximum 10/10 score, the team must provide empirical evidence of outreach reach (impressions, shares, comments), active tagging of the organizing committee, and structured agile collaboration records submitted before the 31st October deadline.

## 2. Four-Phase Campaign Roadmap (Until 31st October)

| Phase & Dates | Strategic Focus | Deliverables & Activities | Target KPIs |
|---|---|---|---|
| **Phase 1: Concept Launch** (Oct 15 – Oct 19) | Publicize contest concept & mission debut. | • Official competition announcement<br>• Contest vision infogram<br>• Mission logo & rover teaser render | 500+ Impressions<br>50+ Engagements<br>10 University Shares |
| **Phase 2: Tech Deep-Dive** (Oct 20 – Oct 24) | Lunar South Pole science & autonomy outreach. | • Carousel: 'Why is Lunar Ice Hard to Find?'<br>• GPR vs Spectrometer explainer<br>• Tag contest organizers | 800+ Impressions<br>150+ Swipes<br>5 Club Reposts |
| **Phase 3: Interactive Demo** (Oct 25 – Oct 28) | Community engagement with 3D digital twin. | • 45s screen capture reel of 3D Payload HUD & Mobile Simulation<br>• Community poll on lunar resources<br>• Live Q&A; in comments | 1,200+ Impressions<br>100+ Poll votes<br>30+ Comments |
| **Phase 4: Final Sprint** (Oct 29 – Oct 31) | Team cooperation showcase & competition eve. | • Team Spotlight: Meet the 4 Engineers<br>• Behind-the-scenes lab/code photos<br>• Final contest tribute post | 1,500+ Total Reach<br>Complete Dossier<br>100% On-Time |

## 3. Communications Specialist Playbook & Role Distribution

Table 6 explicitly identifies the 'Communications Specialist' as a key evaluated element. The Communications Specialist leads external dissemination while each team member contributes domain copy.

| Team Member | Primary Technical Role | Outreach & Communications Responsibility |
|---|---|---|
| **Hajra** | UI / Dashboard Lead | **Lead Communications Specialist:** Manages social postings, graphics generation, community replies, and press communication. |
| **Haider Taqvee** | Sensor Physics Lead | **Science Writer:** Drafts plain-language explanations of GPR dielectric physics and USGS spectroscopy for public posts. |
| **Abdul Wasay** | Machine Learning Lead | **Data Viz Lead:** Generates high-resolution confusion matrices, ROC plots, and benchmark graphics for carousels. |

| | | |
|---|---|---|
| **Saad** | Autonomy Systems Lead | **Documentation Officer:** Maintains meeting minutes, GitHub project board, attendance records, and collaboration proof. |

# 4. Ready-to-Publish Outreach Assets & Copy

## Asset 1: LinkedIn Launch Post Copy (Phase 1)

■ **Exciting News from IST Islamabad!**
Our team is thrilled to participate in ISSSP 2026 Track 3 (Lunar Surface Robotics & Autonomous Exploration)! ■

As humanity prepares to establish sustainable bases on the Moon through Artemis and international lunar research programs, one question remains paramount: How do we locate volatile water ice trapped inside permanently shadowed craters at 40 Kelvin?

Introducing **L-MINER Mk-II:** An Autonomous Multi-Tier Rover for Subsurface Lunar Mineral Classification.
Equipped with 500 MHz Ground Penetrating Radar, 0.5–2.5 µm VNIR-SWIR Reflectance Spectroscopy, and edge Random Forest ML, L-MINER's autonomous payload firmware navigates hazards, monitors power, and maps resources without Earth teleoperation lag.

Proud to represent IST alongside teammates Haider Taqvee, Abdul Wasay, Saad, and Hajra. Follow our journey before Oct 31st!
*#ISSSP2026 #LunarExploration #SpaceRobotics #ISTIslamabad #MoonToMars #ISRU #AutonomousSystems*

## Asset 2: Twitter/X Technical Explainer Thread (Phase 2)

■ **1/5** How does an autonomous science payload 'see' water ice buried 2 meters under lunar dust? Here's how our project for #ISSSP2026 Track 3 works! ■
**2/5** Optical cameras only see the top 10 microns of dust. That's why L-MINER Mk-II combines TWO complementary sensors: an underbelly 500 MHz GPR (for depth) + a robotic-arm VNIR-SWIR Spectrometer (for chemistry). ■■
**3/5** Water ice has a sharp dielectric boundary ($\varepsilon_r \approx 3.1$) vs dry regolith, creating high-amplitude radar echoes. Spectroscopically, it reveals diagnostic absorption troughs at 1.25, 1.5, and 2.0 µm (matching Chandrayaan-1 $M^3$ data)! ❄■
**4/5** Our edge Random Forest model extracts a 10-D feature vector, compressing 32,000+ raw data points into instant classification with 95%+ accuracy in under 15 ms! ■
**5/5** Huge thanks to the ISSSP 2026 organizing committee for inspiring student aerospace innovation! Stay tuned for our 3D Digital Twin demo! ■

# 5. Team Cooperation & Internal Communication Protocol

To provide empirical proof of 'Team cooperation & communication' (Table 6, Item 1), the team instituted a formal Agile/Scrum collaboration framework documented through GitHub, Discord, and Trello.

| Pillar | Tools Used | Operational Protocol & Artifacts |
|---|---|---|
| **Daily Standups** | Discord / Meet | 15-minute daily sync at 18:00 PKT. Agenda: Completed items, blockers, next sprint tasks. Minutes logged. |
| **Version Control** | GitHub Repo | Strict PR reviews. Dedicated modules for data_generator, model, l0_l1_logic, and dashboard. |
| **Sprint Tracking** | Trello Board | Kanban swimlanes tied directly to Table 6 deliverables. Individual assignee tags. |
| **Deliverable Archive** | Google Drive | Centralized repository for poster assets, slide decks, and code builds. |

# 6. Evidence Submission Checklist for Judges (Before 31st Oct)

- **[X] Publicity Posts Live:** Links and screenshots of LinkedIn, Twitter/X, and Instagram posts tagging the competition.
- **[X] Engagement Metric Sheet:** Total views, impressions, reposts, and student comments compiled into an appendix.
- **[X] Team Standup Logs:** 14 consecutive daily meeting minute summaries signed off by all four team members.
- **[X] GitHub Contribution Matrix:** Multi-author commit frequency proving active collaboration across all modules.
- **[X] Official Submission Package:** Uploaded to the competition portal prior to 23:59 PKT on 31st October 2026.